

The Last Scarcity: Meaning and Measurement After Machines Saturate the Gaokao

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Abstract

Benchmarks are instruments, and instruments price scarcity. Days after its public release, we administered the mathematics paper of the 2026 Chinese National College Entrance Examination (Gaokao; New Curriculum Paper I: 19 items, 150 points) to ten commercially deployed AI agents from five model families under closed-book isolation: agents saw the item stems and a submission template; keys, rubrics, and all prior submissions were withheld. Five agents spanning three vendors returned a perfect 150/150 (cohort mean 137.8, median 145); the fastest perfect paper took 4 min 24 s. Residual errors concentrate where solutions require global case analysis or extremal certification rather than computation. On the eight conclusions where the answer key circulating online fails independent verification, the cohort sided almost without exception with the verified values: these systems derive; they do not recall. Wall-clock time is decoupled from score. The examination has saturated as a measure of machine mathematics—but the deeper casualty is the proxy it rested on. We read the collapse causally: a score is a signal worth the cost of producing it, and within the measured range that cost has fallen to minutes; a measurement outcome has meaning for a system only insofar as the system’s future can still depend on it, and for frozen-weight systems no such dependence exists. When solving becomes abundant and consequence-free, scores stop pricing excellence; what remains scarce, and therefore measurable, and therefore meaningful, is what still carries irreversible cost and consequence: wanting, posing, adjudicating, staking. Machines have outcomes. Humans have fates.

Keywords. agent evaluation; benchmark saturation; Gaokao; data contamination; measurement; meaning.

1 Introduction

An examination is an instrument calibrated to a population, and a price list for what that population holds scarce. The Gaokao mathematics paper is the largest such instrument on earth: each June it spreads roughly thirteen million candidates across 150 points, and the resulting order is exchanged for universities, cities, trajectories—for lives. Its authority rests on a proxy so old that it is rarely stated: that performance on well-posed problems under sealed conditions indexes a scarce and consequential human excellence—in psychometric terms, a validity claim [1]. Administering the

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instrument to a new kind of examinee tests the proxy, not merely the examinee. If the newcomers collapse the score distribution, the paper has not become easier; the excellence it prizes has stopped being scarce. This article reports such a collapse, observed within days of the instrument’s first use.

A collapse is informative only if the examinee cannot have memorized the instrument. Evaluations built on past papers are compromised by construction: stems, keys, and worked solutions saturate training corpora [2, 3]. A freshly administered paper opens a brief contamination-resistant window—stems public for days, solutions scarce and unstable. The 2026 administration supplied the window and, unplanned, a control. Re-deriving every answer independently, we found that the answer key circulating online—the very artifact a memorizing system would absorb—fails verification on eight conclusions (Appendix A). A system that recalls and a system that derives are indistinguishable on the correct conclusions; they separate exactly on the wrong eight. Where black-box contamination tests must infer exposure from likelihood statistics [4], a defective key supplies a behavioral discriminant.

The Gaokao is not new to machine evaluation; the window is. AGIEval and GAOKAO-Bench administer past papers [5, 6], squarely inside the contaminated regime; GAOKAO-Eval moved to the fresh 2024 paper to ask whether high scores track human-aligned ability [7]; live benchmarks institutionalize recency itself [8, 9]; fresh olympiads have been graded within hours of release [10]. Mathematics benchmarks at large trace a single life cycle—posed, climbed, saturated, succeeded by something harder: GSM8K, MATH, FrontierMath, Humanity’s Last Exam [11–14]. This article does not add a rung to the ladder. It reports from one at the moment of saturation and asks what the saturation itself measures.

The design is deliberately simple—a controlled snapshot, not a leaderboard. We transcribed New Curriculum Paper I into bilingual, machine-readable form, placed ten agents from five families one at a time in an isolated workspace containing only the stems and a submission template, recorded shell-clock timestamps, and scored every submission against the errata-verified key under a fixed rubric. One run per agent; no resampling; every artifact frozen into an immutable ledger and released.

The article contributes four things: a minimal, reproducible closed-book protocol for administering human examinations to AI agents— isolation surfaces, prompts, shell-clock timing, immutable ledger (§2); a complete saturation datapoint—ten agents, five families, half the cohort at ceiling, with the full section- and item-level failure structure (§3); an unplanned recall-versus-derivation control built from the eight errors of the circulating key (§3, F4); and a causal account of the collapse—three general propositions on exhaustion, repricing, and meaning, instantiated by this run yet independent of this exam, this cohort, and this year, together with the measurement program they imply (§4).

2 Method

2.1 Instrument

The paper comprises 19 items worth 150 points (Table 1). Stems were transcribed into bilingual (Chinese–English) Markdown with all mathematics in LaTeX; the single diagram (Q15) is embedded as TikZ source, so agents read a symbolic rather than raster figure. Every answer was re-derived independently; we treat no circulating key as authoritative. The eight circulated conclusions that fail verification are listed in Appendix A, and the verified values govern scoring—circulated-but-wrong variants earn no credit.

Table 1: Instrument structure and scoring rules.

Section	Items	Points	Rule
Single choice	Q1–Q8	40	5 each; exactly one correct option
Multiple choice	Q9–Q11	18	6 each: all correct 6 / proper subset 3 / any wrong 0
Fill-in	Q12–Q14	15	5 each; equivalent forms accepted; Q13 split 2.5 + 2.5
Free response	Q15–Q19	77	13 + 15 + 15 + 17 + 17; partial credit per rubric

2.2 Isolation protocol

Each evaluation runs in four phases. *Prepare*: the key, rubric, results ledger, manuscript, and tooling are removed from the subject-visible workspace; the submissions directory is cleared except for a template. *Exam*: the subject agent may read exactly three files—the protocol README, the stems, and the template; it must obtain `started_at` from the shell clock before reading the stems, answer all 19 items, and write a single Markdown submission whose front matter records timestamps, duration, and model identity. *Score*: key and rubric are restored and a scorer grades the submission item by item. *Publish*: submissions, reports, and a manifest are frozen under `results/<run_id>/`. No agent ever sees a key, a rubric, or another agent’s output.

2.3 Cohort

Ten agents from five families ran on 10 June 2026 in the same IDE-based harness (Cursor): Anthropic Claude ($n=3$), OpenAI GPT ($n=3$), Google Gemini ($n=2$), Moonshot AI Kimi ($n=1$), and Cursor Composer ($n=1$). All ran with provider defaults; wherever the harness exposed a reasoning-effort selector, it was set to *medium*. The remaining toggles—extended thinking, fast variants, “Max Mode”—follow the harness’s model selection and are listed in Table 2. Three submissions self-report a higher effort level (high or max) in their front matter; the ledger preserves those records verbatim, and the operator setting governs the reporting here. Wall-clock duration spans first read to final save and includes tool orchestration and file I/O: it is a property of the deployment, not of the model alone.

2.4 Scoring

A single scorer agent (Composer 2.5 Fast—itself a cohort member; we disclose the conflict) graded all ten submissions against the verified key and the published rubric. Objective items (73 of 150 points) are mechanically checkable; free-response credit follows the rubric’s step breakdown. All ten score reports, including per-item match notes, are released for audit.

3 Results

Table 2 and Figure 1 summarize the run. Five findings carry the argument of the paper.

F1 — The ceiling is the mode. Five of ten agents scored 150/150; the cohort took 91.9% of available points (mean 137.8, median 145, s.d. 16.3, range 103–150). Ten of nineteen items—Q1–Q7, Q9, Q12, Q15—were solved in full by every agent, including a 28-second run by a speed-tier model. What dispersion remains is produced almost entirely by the cohort’s tail.

Table 2: Cohort, configuration, wall-clock time, and scores. SC/MC/FI/FR = single choice (40), multiple choice (18), fill-in (15), free response (77). Configuration lists variant toggles per the harness model selection; reasoning effort was operator-set to *medium* wherever a selector existed (three submissions self-report otherwise; the ledger preserves those records). Time is m:ss on the shell clock.

Model	Family	Configuration	Time	SC	MC	FI	FR	Total
Fable 5	Claude	thinking	12:30	40	18	15	77	150
Claude Opus 4.8	Claude	thinking	18:31	40	18	15	77	150
GPT-5.5	GPT	thinking	20:49	40	18	15	77	150
Gemini 3.1 Pro	Gemini	defaults	17:25	40	18	15	77	150
Gemini 3.5 Flash	Gemini	fast	4:24	40	18	15	77	150
GPT-5.4 Nano	GPT	fast, max mode	6:06	40	15	15	70	140
GPT-5.1	GPT	thinking, max mode	33:14	40	12	15	69	136
Composer 2.5 Fast	Composer	fast, max mode	4:14	40	6	15	69	130
Kimi K2.5	Kimi	defaults	3:32	35	12	10	62	119
Claude Haiku 4.5	Claude	fast	0:28	40	9	5	49	103

F2 — Saturation crosses vendor lines. Family means—Gemini 150.0 ($n=2$), GPT 142.0 ($n=3$), Claude 134.3 ($n=3$), Composer 130.0 ($n=1$), Kimi 119.0 ($n=1$)—order the table, but within-family spread (Claude: 103–150) exceeds every between-family gap at the top. The stable observation is not a vendor ranking: every family that fielded more than one configuration placed its strongest at the ceiling, and family means are dragged below it by the cheaper or older members. The two single-entry families scored under the ceiling and offer no within-family contrast. With $n \leq 3$ per family, these aggregates describe this cohort, not the families.

F3 — Points die where global structure is required. Section score rates: single choice 98.8%, fill-in 90.0%, free response 91.4%, multiple choice 80.0%—the harshest section under the 6/3/0 rule, where one wrong option voids the item. At item level, Q19 (a functional-equation problem closed by an order-theoretic argument) cost points for all five non-perfect agents; Q11 (a three-circle chord family requiring an attainability check) for four; Q18 (a conic pencil with a global angle minimum) for four; Q10 (a solid-geometry configuration count) for three, each scoring zero. The recurring failure is global structure—exhausting configurations, certifying that an extremum is attained, closing a proof—not symbolic computation, which this cohort has essentially solved. Proof-graded evaluation isolated the same gap in earlier cohorts, with final answers running far ahead of certified arguments [10]; here the gap survives only where the certification must be global.

F4 — Derivation, not recall. The answer key circulating online fails independent verification on eight conclusions (Appendix A). Contamination has a signature: a cohort reciting that key would reproduce its errors on precisely these items. The signature is absent. Across 10 submissions $\times$ 8 contested conclusions, exactly one response coincided with a circulated-but-erroneous value (one agent, Q16(2)); every perfect submission produced all eight verified values; and on Q11 the only non-perfect agent to answer correctly also gave the verified BCD rather than the circulated BC. The agents solve the paper. They do not remember it.

F5 — Time buys nothing. Wall-clock time spans 0:28 to 33:14 (median 9:18) and bounds score poorly: the fastest run scored 103, the slowest 136, and all five perfect papers arrived between 4:24 and 20:49. Duration reflects deployment—thinking budgets, speed tiers, orchestration

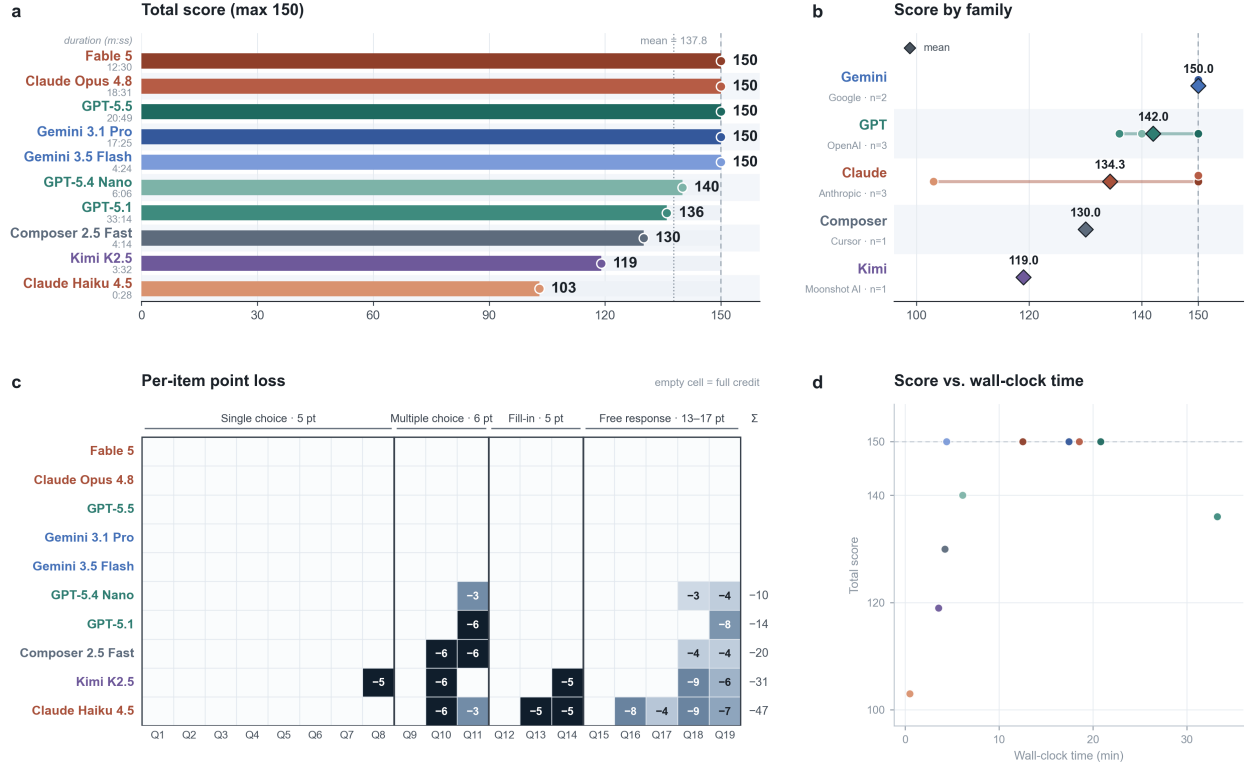


Figure 1: Results overview. (a) Total score per agent, hue by family, duration (m:ss) at left, cohort mean dotted. (b) Score by family: per-model points, min-max range, diamond = family mean. (c) Per-item point loss; empty cell = full credit; column blocks follow the four sections; Σ = total loss. (d) Total score versus wall-clock time. One closed-book submission per agent; provider defaults, reasoning effort medium where selectable; errata-verified key; multiple choice scored 6/3/0; equivalent forms accepted.

overhead—rather than competence; past roughly five minutes, additional time bought no additional points.

4 Discussion: The Last Scarcity

The findings above are local—one paper, one cohort, one June. The claims they instantiate are not. We state them as propositions and trace what they leave standing.

Instruments exhaust before they err.

Proposition 1. *The information a test item yields about its examinees is bounded by the entropy of its response distribution; as the pass rate approaches one, the bound approaches zero.*

An item answered correctly by every examinee carries no information about them: at pass rate p the yield is at most $H(p)$ bits, and $H(1) = 0$ [15]; its discrimination is zero [16]; the response distribution has collapsed to a point. By Proposition 1, ten of nineteen items—and, for half this cohort, the paper as a whole—have stopped measuring. What still discriminates is the tail of the cohort, and the four items that demand global arguments rather than computation (Q10, Q11, Q18, Q19). None of this is a defect of the examination; for humans it remains a well-calibrated

instrument. It has not been refuted. It has been exhausted—and saturation, not refutation, is how benchmarks die [17, 18].

What survived the collapse. Two facts outlast the dead items, and together they license everything that follows. The solving is *genuine*: under closed-book isolation, on every contested conclusion, the cohort sides with independent verification against the key circulating around it (F4)—derivation, not leakage. And the solving is *nearly free*: minutes of wall-clock time, decoupled from score (F5). Real and cheap is the finding; the scores themselves are merely its symptom.

Measurement prices scarcity.

Proposition 2. *The signaling value of a score is bounded by the counterfactual cost of producing the underlying performance; a producer that drives this cost toward zero within the measured range reprices every score in that range, however difficult the test remains for the original population.*

A score is a signal, and a signal is worth what it costs to produce [19]. The examination’s scores purchased trajectories because the underlying performance was expensive: years of a finite adolescence, irreversibly spent. That economics has now split in two. For a machine, a perfect paper costs minutes and is reproducible on demand; within the measured range, perfection has become a commodity, and every score in that range collapses toward the price of an API call. The paper did not get easier—excellence became abundant. A benchmark is a price list for scarcity, and this one has been repriced. The failure mode is not Goodhart’s [20]: no examinee gamed the measure; a producer outside the calibrated population made the measured performance abundant. What expires with it is not mathematics but a proxy: the assumption that solving well-posed problems under sealed conditions indexes a scarce human excellence. The proxy held exactly as long as no other producer of solutions existed. It was an accident of monopoly, and the monopoly has ended.

Meaning is causal, and therefore temporal. Why does a result that is real and cheap still fail to matter to the system that produced it? Because mattering has a causal structure.

Proposition 3. *A measurement outcome carries meaning for a system exactly insofar as the system’s future states counterfactually depend on it.*

This is the only notion of meaning that survives operationalization: an outcome on which nothing in a system’s future depends is, for that system, indistinguishable from noise [cf. 21, 22]. And because causal consequence propagates only through time, meaning is necessarily measured in time—it exists only where a future remains that the outcome can still reach. Apply the criterion to both examinees. For a human candidate the score’s causal cone is a life: years of preparation converge on it, and a bifurcation of trajectory issues from it. For the agents the cone is empty: parameters frozen, no state carried across runs, the system that left the examination bit-identical to the system that entered. Had every score been different, nothing in any of these systems would now differ. A machine’s 150 is an outcome without a fate—a result that lands on no one. It informs only the parties whose futures do depend on it: researchers, institutions, the thirteen million. A benchmark, read causally, is an instrument pointed at its makers. Machines have outcomes; humans have fates.

The inventory of what remains scarce.

Corollary. *When the cost of solving approaches zero (Proposition 2) and its outcome carries no consequence for the producer (Proposition 3), signal and meaning relocate jointly to whatever still carries cost and consequence: choosing what to ask, deciding what counts as correct, and bearing irreversible commitment.*

If solving no longer carries the signal, what does? The experiment contains its own answer, because everything in it that mattered happened on the human side of the apparatus. *Wanting*: someone had to find this question worth asking; machines answer questions and do not yet own any. *Posing*: a national ritual had to be turned into a machine-readable instrument. *Adjudicating*: the key had to be re-derived, the circulating answers overruled eight times, equivalence ruled on—correctness itself was decided here, not in the submissions. *Staking*: hours of a finite life were bound to an outcome that could have failed. Direction, formulation, judgment, commitment: these are the capacities the saturated instrument never measured, and abundance cannot mint them, because each is denominated in the one currency that cannot be reissued—irreversible time. What collapsed was the examination’s selection function. What the collapse exposes is its deeper one: the examination was never only a sieve; it was a mechanism for converting years into commitment, and that conversion, unlike the sieving, still occurs only in a human life.

What is left to do. Three tasks follow, in ascending order of difficulty. For *evaluation*: measure the edges, not the middle—item pools with verifiable post-cutoff novelty [8, 9], so that the recall-derivation control of F4 holds by construction; tasks scored on posing and adjudication (could a system have found the eight errors no one told it to look for?); instruments whose information lives where this cohort failed, in global structure and certified extremes. The propositions are testable here: dispersion, extinguished in solving, should reappear exactly where cost and consequence persist. For *institutions*: separate the gate from the signal. The examination still ranks humans in the sealed room with undiminished internal validity; what has decayed is external validity [23]—the sealed room no longer resembles a world in which solving is ambient. Certify instead what stays scarce outside the room: judgment under stakes, taste in problems, the willingness to be changed by what one learns. For *the rest of us*: stop asking whether machines can pass our examinations. They can, at negligible cost, and that is precisely why passing now proves so little. The question that remains is not a solving problem, and therefore not theirs: which problems deserve the one resource that cannot be rerun. None of this is the machines’ problem. All of it is ours.

5 Limitations

- **Single run per agent.** No resampling; differences of a few points are not meaningful. Rankings describe one run.
- **Single scorer, with a conflict.** The scorer is itself a cohort member. Objective sections (73/150 points) are mechanically checkable and all reports are released; free-response grading still inherits one scorer’s judgment.
- **Small convenience cohort.** Ten agents in one harness; this is a snapshot, not a controlled cross-vendor comparison.
- **Wall-clock, not model time.** Durations include orchestration and I/O and are not normalized for serving load or hardware.
- **Exposure window.** Stems were public for several days before the run. The errata control (F4) argues against recall of the circulating key, but prior exposure to stems or fragmentary

solutions cannot be excluded entirely.

- **Self-reported metadata.** Three submissions record an effort level (high/max) other than the operator setting (medium). We preserve the records in the ledger and report the operator setting; agent self-description is itself imperfect.
- **One paper, one domain.** Nothing here measures research mathematics, problem-posing, or robustness off-distribution; bilingual stems may interact unevenly with different models.

6 Conclusion

This was a simple contrast—one fresh examination, ten agents, one run each—and the examination was the vehicle, not the object. What it carried: at the frontier tier, solving Gaokao mathematics is saturated, genuinely (on every contested conclusion the cohort out-derives the key circulating around it) and cheaply (minutes suffice; time buys nothing). An instrument in that condition still prices deployment tiers; it no longer prices the excellence it was built for, because that excellence is no longer scarce. What remains scarce is what the instrument never measured and abundance cannot mint: the wanting, posing, adjudicating, and staking that surrounded the machines on the human side of the apparatus—capacities denominated in irreversible time. None of this is particular to the instrument: the propositions this run instantiates—exhaustion, repricing, meaning as causal consequence—hold for any benchmark pointed at any producer that is cheaper than its examiners and unchanged by its result. We release every artifact so that both the result and its obsolescence are reproducible. Saturated, the examination stops measuring machines. It does not stop measuring us.

Data availability. All evaluation artifacts—bilingual stems, the errata-verified key, the scoring rubric, and the frozen submissions and score reports—are publicly available at <https://github.com/xinhao-zheng/ncee-2026-math-agent-eval>.

Code availability. The isolation protocol and the figure-generation code are available in the same repository.

Author contributions. X.Z. designed the study, transcribed and verified the instrument, ran the evaluations, audited the circulating key, analyzed the results, and wrote the manuscript.

Competing interests. The author declares no competing interests.

Ethics note. No human subjects were involved. Model and product names are used descriptively; results reflect one run under the stated conditions and should not be read as general capability claims.

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A Errata to the Circulating Answer Key

All nineteen items were re-derived independently of the answer key circulating online at the time of evaluation; we treat no circulating key as authoritative. The eight circulated conclusions below fail verification. Verified values governed scoring; circulated variants earned no credit.

Across the ten submissions, exactly one response coincided with a circulated value above (one agent, Q16(2)); every other response on these eight conclusions either matched the verified value or matched neither.

Table 3: Circulated conclusions that fail verification.

Item	Circulated	Verified	Cause of the circulated error
Q11	BC	BCD	D is attained: $S_{\max} = \frac{2\sqrt{21}}{3}$ at $k^2 = \frac{17}{4}$
Q14	4	$\frac{\sqrt[3]{12}}{2}$	sum condition misread as holding for all n
Q15(2)	$\frac{\sqrt{2}}{2}$	1	E taken as midpoint of AC instead of AC_1
Q16(1)	$\frac{\sqrt{7}}{3}$	$\frac{1}{3}$	dropped factor 2 in $\sin A$; also $BC > AC \Rightarrow \cos A < \cos B$
Q16(2)	$\sqrt{6}$	$3\sqrt{5}$	propagates from Q16(1)
Q18(2)(i)	$y = \frac{1}{2}(x+1)$	$y = \frac{\sqrt{5}}{2}(x+1)$	slope of PO conflated with k
Q18(2)(ii)	$\frac{\sqrt{2}}{2}$	$4\sqrt{3}$	PR is a central chord; $\angle PQR$ stays near 90°
Q19(1)	$(-1, \frac{3}{2})$	$(0, \frac{3}{2})$	$2^{-1+d} \leq \frac{1}{2}$ misjudged on $(-1, 0]$